

Applied Statistics (MATH10096)

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Checking the model assumptions: residuals

↪ The assumptions of the linear model are:

- 1 The structural part of the model, $\mathbb{E}(\mathbf{y}) = \mathbf{X}\beta$, is correct. Note that this assumes $\mathbb{E}(\epsilon) = \mathbf{0}_n$. This assumption implies that the expectation includes all the important predictor variables in the precisely correct functional form.
- 2 The errors have constant variance (*homoscedasticity*).
- 3 The errors are independent.
- 4 The errors follow a normal distribution.

Checking the model assumptions: residuals

- ↪ Regarding assumption 1., as we learned in Section 3, nonlinear effects of predictors can be accommodated within the framework of linear models; the only requirement is that the resulting model is linear in the parameters.
- ↪ Regarding assumption 4., if the normality assumption does not hold, tests and confidence intervals are not exact.
- ↪ However, we can appeal to the central limit theorem, which ensures that the tests and confidence intervals become increasingly accurate approximations as the sample size grows.
- ↪ Hence, we can afford to ignore the issue, provided the sample size is sufficiently large or the violation is not particularly severe (i.e., the distribution is not heavy-tailed, heavily multimodal, or heavily skewed).

Checking the model assumptions: residuals

↪ The errors are not observable but we can examine the residuals

$$\hat{\epsilon} = \mathbf{y} - \hat{\boldsymbol{\mu}}, \text{ where } \hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

↪ The residuals contain all the information left in the data after fitting the model.

↪ Note that, provided that $\mathbb{E}(\epsilon) = \mathbf{0}_n$,

$$\mathbb{E}(\hat{\epsilon}) = \mathbb{E}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}\boldsymbol{\beta} = \mathbf{0}_n$$

↪ To evaluate the covariance matrix of $\hat{\epsilon}$ we will make use of the influence matrix $\mathbf{A} = \mathbf{Q}\mathbf{Q}^T$. Recall that $\mathbf{A} = \mathbf{A}^T$ and $\mathbf{A}\mathbf{A} = \mathbf{A}$.

↪ Further note that we can write the residuals as

$$\hat{\epsilon} = \mathbf{y} - \hat{\boldsymbol{\mu}} = \mathbf{y} - \mathbf{A}\mathbf{y} = (\mathbf{I}_n - \mathbf{A})\mathbf{y}.$$

↪ We therefore have (under the assumption of independent error terms with constant variance σ^2)

$$\mathbf{V}_{\hat{\epsilon}} = \text{var}\{(\mathbf{I}_n - \mathbf{A})\mathbf{y}\} = (\mathbf{I}_n - \mathbf{A})\text{var}(\mathbf{y})(\mathbf{I}_n - \mathbf{A}) = (\mathbf{I}_n - \mathbf{A})\sigma^2.$$

Checking the model assumptions: residuals

- ↪ Further, under the model assumptions, $\hat{\epsilon}$ is a linear transformation of the multivariate normal random vector ($\mathbf{y}$), so it must have a normal distribution

$$\hat{\epsilon} \sim N(\mathbf{0}, (\mathbf{I}_n - \mathbf{A})\sigma^2).$$

- ↪ We observe that although the errors may have equal variance and be uncorrelated, the residuals do not.
- ↪ Specifically,

$$\text{var}(\hat{\epsilon}_i) = (1 - A_{ii})\sigma^2, \quad \text{and} \quad \text{cov}(\hat{\epsilon}_i, \hat{\epsilon}_j) = -A_{ij}\sigma^2, \quad i \neq j,$$

- ↪ Here A_{ii} is the i th diagonal element of $\mathbf{A}$, and A_{ij} is its (i, j) th element.

Checking the model assumptions: residuals

- ↪ We cannot do anything about the slight lack of independence, but the residuals can be standardized to have constant variance, by defining

$$\tilde{\epsilon}_i = \frac{\hat{\epsilon}_i}{\sqrt{1 - A_{ii}}} = \frac{y_i - \hat{\mu}_i}{\sqrt{1 - A_{ii}}}, \quad i = 1, \dots, n,$$

- ↪ Note that

$$\text{var}(\tilde{\epsilon}_i) = \text{var}\left(\frac{\hat{\epsilon}_i}{\sqrt{1 - A_{ii}}}\right) = \frac{1}{1 - A_{ii}} \text{var}(\hat{\epsilon}_i) = \frac{(1 - A_{ii})\sigma^2}{1 - A_{ii}} = \sigma^2.$$

- ↪ Further, note that $\tilde{\epsilon}_i/\sigma \sim N(0, 1)$ if the model assumptions hold, and the correlation between $\tilde{\epsilon}_i/\sigma$ and $\tilde{\epsilon}_j/\sigma$, for $i \neq j$, tends to be small.

Checking the model assumptions: residuals

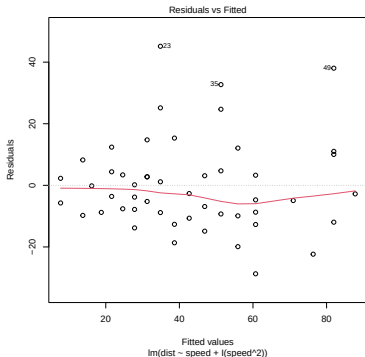
- ↪ The best way to check residuals is to examine a variety of diagnostic plots, always looking for evidence of any violation of the linear modelling assumptions.
- ↪ Let us revisit the `cars` dataset and consider the running example model

$$\text{dist}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2).$$

```
cm <- lm(dist ~ speed + I(speed^2), data = cars)
```

Checking the model assumptions: residuals

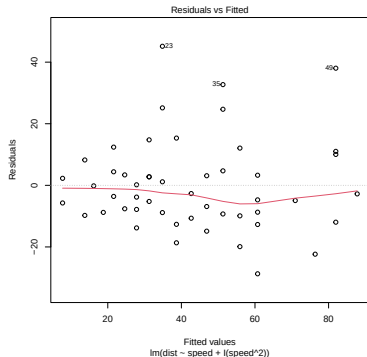
```
plot(cm, which = 1)
```



- This plot shows the model residuals, $\hat{\epsilon}_i$, against the model fitted values, $\hat{\mu}_i$.
- The residuals should be evenly scattered above and below zero, independent of the x-axis positions.
- The red line should be constant around zero.
- A trend in the mean of the residuals is a sign of remaining systematic structure and usually results from an erroneous model structure (assumption 1.): e.g., assuming a linear relationship with a predictor, when a quadratic is required, or omitting an important predictor variable.

Checking the model assumptions: residuals

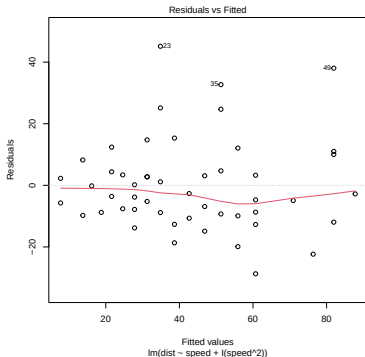
```
plot(cm, which = 1)
```



- A plot of the residuals against the predictors and potential predictors can help in further investigating/diagnosing the misspecification of the model mean structure.
- For plots of the residuals against the predictors not included in the model, any observed structure may indicate that this predictor should be included in the model.

Checking the model assumptions: residuals

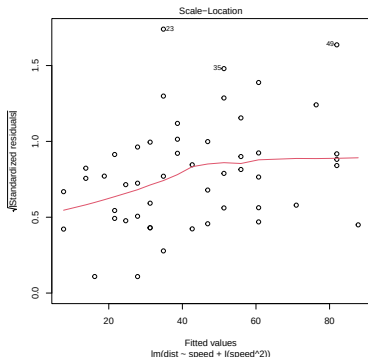
```
plot(cm, which = 1)
```



- This plot also allows investigating the assumption of constant variance.
- A trend in the variability of the residuals (e.g., a funnel shape) suggests that the variance of the response is related to its mean, violating the constant variance assumption.
- Transformation of the response or use of a generalised linear model may help, in such cases.
- The plot shown shows some indication of increasing variance with mean, which would somewhat violate the constant variance assumption, although the effect is not extreme here.

Checking the model assumptions: residuals

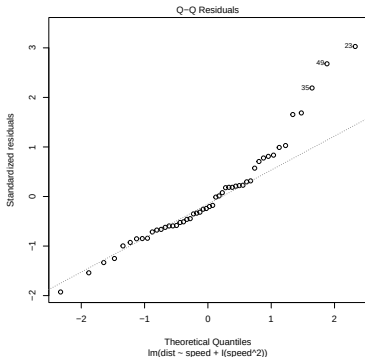
```
plot(cm, which = 3)
```



- This plot is a scale-location plot. The raw residuals are standardized by dividing by their estimated standard deviation, $\hat{\sigma} \sqrt{1 - A_{ii}}$.
- The square root of the absolute value of each standardized residual is then plotted against the corresponding fitted value.
- It can be easier to judge the constant variance assumption from such a plot, and the square root transformation reduces the skew in the distribution, which would otherwise be likely to occur.
- We expect the red line to be almost constant about one.
- There is some suggestion of variance increasing with mean in this plot, but the effect is not very large, so is unlikely to cause a big problem.

Checking the model assumptions: residuals

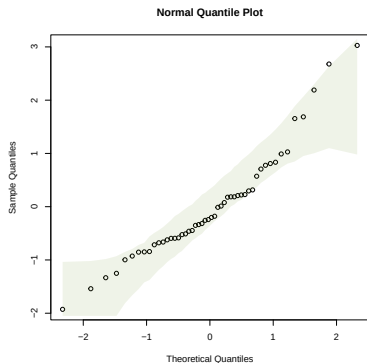
```
plot(cm, which = 2)
```



- This plot is a normal Q-Q (quantile-quantile) plot.
- The standardized residuals are sorted and then plotted against the quantiles of a standard normal distribution.
- If the residuals are normally distributed then the resulting plot should look like a straight line relationship, perturbed by some random scatter.
- In this case the few large residuals show signs of being a bit too big to have come from the same normal distribution as the others, but again, the effect is not large relative to what can occur by chance even when the model assumptions are fine.

Checking the model assumptions: residuals

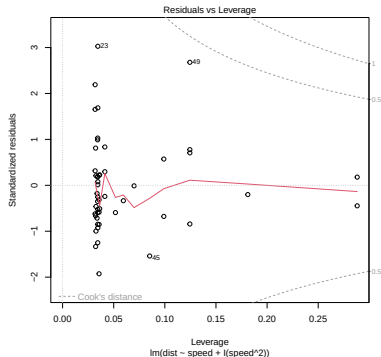
```
library(ecostats) qqenvelope(cm)
```



→ Due to the difficulty in judging what deviation of a straight line can still be tolerated, there are alternative implementations which also plot a corresponding envelope, giving an idea of the natural variation.

Checking the model assumptions: residuals

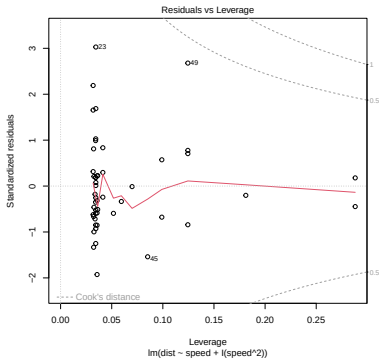
```
plot(cm, which = 5)
```



- This graphic plots the standardized residuals against the **leverage** of each datum.
- The leverage is simply A_{ii} , which measures the potential for a particular datum to influence the overall model fit.

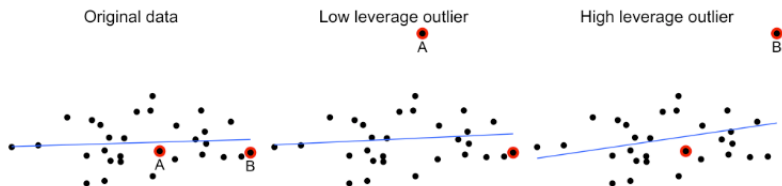
Checking the model assumptions: residuals

```
plot(cm, which = 5)
```



- The trace of the matrix $\mathbf{A}$ is p . The dimension of the model space is also p , and so large leverage is taken to be greater than two or three times the average p/n .
- The combination of a large residual and high leverage implies that the corresponding datum has a substantial influence on the overall fit.
- Large leverage with a small residual generally implies that, although the datum would exert undue influence on the fit, if out of line with the other data, it is actually consistent with the other data, and does not have too great an influence.
- Similarly a large residual may not have much influence on the fit, if the corresponding datum has low leverage.

Checking the model assumptions: residuals



Checking the model assumptions: residuals

- ↪ A good summary of how much influence each datum actually has is provided by its **Cook's distance**.
- ↪ If $\hat{\mu}_i^{[k]}$ is the i^{th} fitted value when the k^{th} datum is omitted from the fit, then Cook's distance is

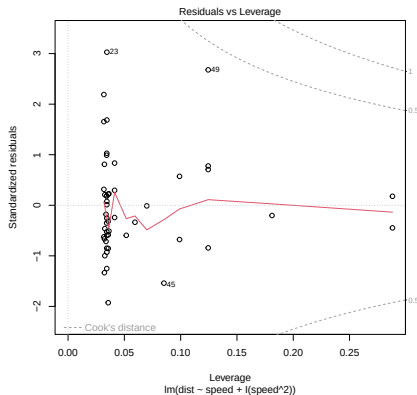
$$d_k = \frac{1}{(p+1)\hat{\sigma}^2} \sum_{i=1}^n (\hat{\mu}_i^{[k]} - \hat{\mu}_i)^2.$$

- ↪ A very large value of d_k indicates a point that has a substantial influence on the model results.
- ↪ Cook's distances above 0.5 are considered borderline problematic, whereas values over 1 are usually considered highly influential.

Checking the model assumptions: residuals

- Cook's distance can be shown to be a function of leverage and standardized residual, so contours of Cook's distance are shown on the plot.

```
plot(cm, which = 5)
```



Checking the model assumptions: residuals

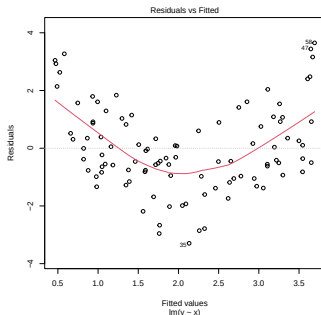
- ↪ We now address the issue of **correlated errors**, which is generally difficult to check because there are many possible correlation patterns that may occur.
- ↪ However, some types of data have a structure that suggests where to look for problems.
- ↪ For instance, data collected over time may exhibit some correlation in successive errors, whereas spatial data may show correlation in the errors of nearby measurements.
- ↪ The presence of strong dependence implies that the data contain less information than the sample size may suggest.
- ↪ In the case of temporal data, one possible approach to detect serial correlation is to plot the residuals against time and check for patterns—any visible pattern is a sign of correlation.
- ↪ An alternative approach is to plot successive pairs of residuals. If temporal correlation is present, methods developed for time series analysis may be used; however, this is beyond the scope of this course.

Checking the model assumptions: residuals

Residual plots: common problems

- Let us simulate data from a quadratic model, fit a linear model and inspect how the residuals look like.

```
set.seed(2)
n <- 100
x <- runif(n, -1, 1)
eps <- rnorm(n)
y <- 1 + 1.5*x + 3*x^2 + eps
fit_wrong <- lm(y ~ x)
plot(fit_wrong, which = 1)
```

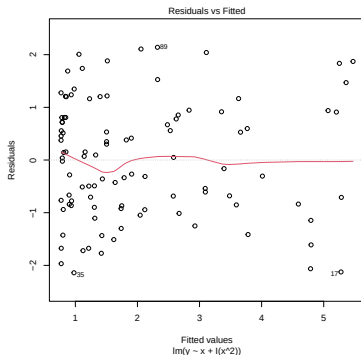


Checking the model assumptions: residuals

Residual plots: common problems

↪ Let us now fit a quadratic model and check the residuals from the resulting fit.

```
fit_correct <- lm(y ~ x + I(x^2))  
plot(fit_correct, which = 1)
```

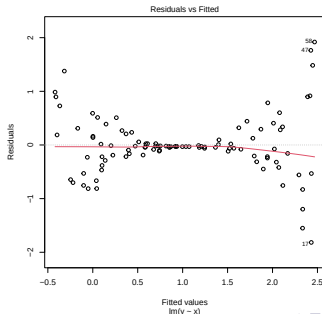


Checking the model assumptions: residuals

Residual plots: common problems

- Let us now simulate data where the variance depends on the predictor (heteroscedasticity) and check how the residual plots look like.

```
set.seed(2)
n <- 100
x <- runif(n, -1, 1)
eps <- rnorm(n, 0, x^2)
y <- 1 + 1.5*x + eps
fit_wrong <- lm(y ~ x)
plot(fit_wrong, which = 1)
```



Checking the model assumptions: residuals

Residual plots: common problems

- Let us now simulate data where the variance depends on the predictor (heteroscedasticity) and check how the residual plots look like.

```
set.seed(2)
n <- 100
x <- runif(n, -1, 1)
eps <- rnorm(n, 0, x^2)
y <- 1 + 1.5*x + eps
fit_wrong <- lm(y ~ x)
plot(fit_wrong, which = 3)
```

